Synthetic American Option Pricing via Jump-HMM-Driven Heston Implied Volatility

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Abstract

Generating realistic synthetic option prices requires implied volatility (IV) as an input, yet IV itself is derived from observed option prices, creating a circular dependency that limits synthetic data availability for machine-learning and risk-analysis applications. We broke this circularity by developing a three-stage pipeline in which IV emerged as an output of a structural model of equity returns rather than as an exogenous input. The pipeline coupled a Jump Hidden Markov Model that generated multi-asset price paths with heavy tails, volatility clustering, and cross-asset tail dependence; a modified Heston stochastic variance process that converted those price paths into implied volatility, where the mean-reversion target was driven by the HMM regime state, days to expiration, moneyness, and an aggregate market mood indicator derived endogenously from the fraction of tickers occupying tail states; and a Cox-Ross-Rubinstein binomial tree that priced American options with early exercise using the resulting IV. A central design choice was initializing the variance process at its mean-reversion target for each strike-expiration pair, so that the smile, skew, and term structure of the IV surface emerged automatically without external calibration data. We calibrated a smile and term-structure function on semiconductor option chains spanning multiple expirations and demonstrated that the smile shape generalized across tickers with only a scalar level adjustment. The multi-expiration calibration revealed a U-shaped ATM term structure driven by gamma compression at short maturities and uncertainty premia at longer horizons, which the model captured through a quadratic extension to the log-DTE term. The framework was implemented as an open-source Julia package designed to generate synthetic data for downstream machine-learning and risk-analysis applications.

1 Introduction

Pricing American-style options requires implied volatility (IV) as an input, yet IV itself is an empirical quantity extracted from observed option prices Black and Scholes [1973]. This circularity poses a practical problem for synthetic data generation: existing approaches either take IV from the market and feed it into a pricing model, or generate price paths without a consistent IV surface. Parametric IV models such as SABR Hagan et al. [2002] and the SVI parameterization Gatheral [2004] treat the IV surface as an object to be fitted to market quotes, not generated from first principles; they produce smooth interpolations of observed smiles but cannot generate new IV dynamics for unobserved market conditions. Local volatility models Dupire [1994], Derman and Kani [1994] recover a deterministic volatility surface consistent with a single cross-section of option prices, yet the resulting surface is static and cannot produce the stochastic IV fluctuations observed in real markets. Stochastic volatility models such as Heston Heston [1993] generate realistic variance paths, but calibrating them requires observed option prices or IV as input, reintroducing the very circularity

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that the synthetic generator must avoid. The result is a gap in the literature: no existing framework generates self-consistent IV surfaces as an emergent output of a structural model of equity returns, without requiring observed option data as a calibration target during simulation.

This gap has practical consequences for a growing set of downstream applications. Empirical studies of IV surface dynamics Cont and da Fonseca [2002] have established that implied volatility exhibits rich cross-sectional and temporal structure, including smile curvature, skew, and term-structure effects that interact with market regimes. Reproducing this structure in synthetic data is essential for training machine-learning models that must generalize across market conditions, for backtesting hedging strategies such as deep hedging Buehler et al. [2019], and for conducting scenario-based risk analysis that requires consistent option prices across strikes and expirations. Historical resampling preserves realized distributional features but cannot produce out-of-sample tail scenarios consistent with a structural model of returns; generative approaches based on neural networks have demonstrated promise for equity price paths but do not directly produce self-consistent option price and IV pairs. The scarcity of realistic synthetic option data therefore remains a binding constraint on progress in computational finance.

In this study, we addressed this gap by developing a framework in which IV was an *output* of a structural model of equity returns, not an input. The pipeline coupled a Jump Hidden Markov Model (JumpHMM) Alswaidan and Varner [2025] that produced multi-asset price paths with realistic heavy tails, negligible linear autocorrelation, and persistent volatility clustering Cont [2001], with cross-asset dependence imposed via a Student-t copula Demarta and McNeil [2005]; a modified Heston stochastic variance process Heston [1993] that converted those price paths into IV paths, where the mean-reversion target θ was a function of the HMM regime state, days to expiration (DTE), moneyness, and an aggregate market mood indicator; and a Cox-Ross-Rubinstein (CRR) binomial tree Cox et al. [1979] that converted the pre-computed IV into American option prices with early exercise. The central innovation was a hybrid θ-function that linked the Heston mean-reversion target to the JumpHMM regime state, producing regime-dependent volatility dynamics. An equilibrium initialization $v_0 = \theta(t=0)$ gave each (ticker, strike, DTE) triple its own initial IV, so the smile, skew, and term structure emerged automatically without external IV data. We calibrated the smile and term-structure function on real semiconductor option chains across multiple expirations, demonstrated that the smile shape generalized across tickers with only a scalar level adjustment per ticker, and showed that the framework produced realistic forward IV paths and American option prices with an endogenous crash-protection premium. The framework is implemented as an open-source Julia package designed to serve as a synthetic option market generator for downstream applications.

2 Background

The framework built on three established models, each contributing a distinct component of the synthetic option generation pipeline: the JumpHMM produced realistic multi-asset price paths, the Heston stochastic variance process provided a mechanism for converting price dynamics into implied volatility, and the CRR binomial tree translated IV into American option prices with early exercise.

2.1 Jump Hidden Markov Model

The Jump Hidden Markov Model (JumpHMM) Alswaidan and Varner [2025] was developed for generating synthetic equity time series that simultaneously preserved three stylized facts of financial returns: heavy tails, negligible linear autocorrelation, and persistent volatility clustering Cont [2001]. The model discretized continuous excess growth rates into N states via equal-probability quantile bins of a fitted Laplace distribution. The hidden state $S_t \in \{1, \ldots, N\}$ at each timestep governed the distribution of the excess growth rate G_t; each state k emitted observations from a location-scale Student-t distribution: $G_t \mid S_t = k \;\sim\; \mu_k + \sigma_k \cdot t_\nu$, where μ_k and σ_k were learned per-state location and scale parameters and ν was the shared degrees-of-freedom parameter. At each timestep, with probability ϵ a Poisson(λ)-distributed jump forced the chain into tail states (bottom or top N_{tail} states), creating sustained dwell periods in extreme-volatility regimes that produced the volatility clustering characteristic of empirical return series. For multi-asset generation, cross-asset dependence was imposed via a Student-t copula Demarta and McNeil [2005] that preserved each asset's marginal distribution while introducing realistic tail dependence; this copula structure ensured

that joint tail events, in which multiple assets simultaneously occupied extreme states, occurred more frequently than a Gaussian copula would imply.

2.2 Heston stochastic variance model

The Heston model Heston [1993] specified the instantaneous variance v of the underlying asset as a mean-reverting square-root process:

$$dv = \kappa(\theta - v)\,dt + \sigma_v \sqrt{v}\,dW_v \tag{1}$$

where κ was the mean-reversion speed controlling how quickly the variance returned to its long-run level, θ was the long-run variance target, and σ_v was the vol-of-vol governing the amplitude of stochastic fluctuations in v. In the standard formulation, the asset price and variance processes were coupled through a correlation ρ between their driving Brownian motions, producing the leverage effect in which negative returns coincided with rising volatility. The key property for our purposes was mean-reversion: the variance process was pulled toward θ, so by controlling θ we could control the equilibrium level of implied volatility. In the standard model θ was a constant; our extension made θ time-varying and state-dependent, linking it to the JumpHMM regime state, contract characteristics, and market conditions (Section 3.2).

2.3 CRR binomial tree

The CRR model Cox et al. [1979] priced options on a recombining binomial lattice with up/down factors $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$ and risk-neutral probability $p = (e^{r\Delta t} - d)/(u - d)$. American options were priced by backward induction with an early-exercise check at each node, making the CRR tree a standard tool for contracts where closed-form solutions do not exist. The CRR tree was a natural choice when IV was provided exogenously, as it required only a single volatility input per contract and handled early exercise without the need for Monte Carlo simulation or PDE solvers. Convergence to the continuous-time Black-Scholes price was guaranteed as the number of tree steps increased, and the computational cost scaled linearly in the number of steps, making the tree efficient for pricing large cross-sections of contracts across multiple strikes and expirations.

3 Methods

3.1 Pipeline overview

The generator produced synthetic American option prices through a three-stage pipeline:

$$\begin{aligned}&\underbrace{\text{JumpHMM(prices)}}_{\text{Stage 1}} \;\rightarrow\; \{S_t,\, s_t,\, \text{jumps}\} \;\rightarrow\; \underbrace{dv\ \text{process}}_{\text{Stage 2}} \;\rightarrow\; \sigma_{\text{imp}} = \sqrt{v_t}\\ &\rightarrow\; \underbrace{\text{CRR}(S_t,\, \sigma_{\text{imp}},\, K,\, \text{DTE},\, r_f)}_{\text{Stage 3}} \;\rightarrow\; P_{\text{american}}\end{aligned} \tag{2}$$

This ordering was dictated by data flow: the JumpHMM produced the price paths and regime-state sequences that the Heston process required as inputs, and the Heston process produced the IV values that the CRR tree required for pricing. Each stage was self-contained, so the pipeline could be run forward without iteration or feedback from downstream stages.

3.2 Modified Heston variance process

We modified the standard Heston process (1) by making the mean-reversion target time-varying:

$$dv = \kappa\big(\theta(t) - v\big)\,dt + \sigma_v \sqrt{v}\,dW_v \tag{3}$$

The mean-reversion target decomposed into a regime-dependent base level, a market mood modulator, and a contract-specific smile and term-structure adjustment:

$$\theta(t) = \theta_{s_t} \cdot \big(1 + \gamma \cdot M_t\big) \cdot \psi(\text{DTE}_t,\ K/S_t) \tag{4}$$

where θ_{s_t} was a learned variance level for HMM state s_t (one value per state, $s_t \in \{1, \ldots, N\}$); $\gamma \geq 0$ was the market mood sensitivity controlling how strongly aggregate stress elevated IV; $M_t \in [0, 1]$

was the aggregate market mood; and $\psi(\mathrm{DTE}, K/S)$ was the smile and term-structure adjustment. The regime-dependent base θ_{s_t} linked the Heston process directly to the JumpHMM state sequence: when the HMM transitioned into a tail state (via a Poisson jump or an extreme Markov transition), θ_{s_t} increased, elevating the IV target. This mechanism produced the leverage effect as an emergent property of the return dynamics rather than as a separate model component: large negative returns drove the HMM into low-numbered tail states with $p_{\text{neg}} \approx 0.52$, which in turn elevated θ.

The function ψ captured the shape of the IV surface with five parameters:

$$\psi(\tau,\ m) = \exp\Big(\beta_1 \ln\tau + \beta_2 \ln m + \beta_3 \ln\tau\ \ln m + \beta_4(\ln m)^2 + \beta_5(\ln\tau)^2\Big) \tag{5}$$

where $\tau = \max(\mathrm{DTE}, 1)$ and $m = K/S$. The parameter β_1 controlled the linear term-structure decay; β_2 controlled the skew (negative values produced put skew, with OTM puts having higher IV than OTM calls); β_3 captured the DTE-skew interaction (the skew flattened at longer maturities); β_4 controlled the smile curvature; and β_5 controlled the DTE curvature, capturing the U-shaped ATM term structure where short-dated and long-dated IV were both elevated relative to intermediate maturities.

Rather than specifying a separate initial variance v_0, we initialized the variance process at its mean-reversion target:

$$v_0 = \theta(s_0,\ \mathrm{DTE},\ K/S_0,\ M_0) \tag{6}$$

This ensured that each (ticker, strike, DTE) triple started at its own equilibrium implied volatility. The initial IV surface inherited the smile, skew, and term structure encoded in ψ without requiring any external IV data. Different contracts on the same underlying received different v_0 values, and the variance process immediately began mean-reverting around its local $\theta(t)$. The value of this design choice lay in eliminating v_0 as a free parameter: the model did not require a separate calibration of the initial IV surface, as it was determined by construction from θ.

The market mood M_t was defined as the fraction of tickers currently occupying tail HMM states, where N_{tail} denoted the number of extreme states at each end of the state space (inherited from the JumpHMM fit):

$$M_t = \frac{1}{N_{\text{tickers}}} \sum_{i=1}^{N_{\text{tickers}}} \mathbf{1}\Big\{ s_t^{(i)} \le N_{\text{tail}} \text{ or } s_t^{(i)} > N - N_{\text{tail}} \Big\} \tag{7}$$

This signal was fully endogenous to the JumpHMM copula model; it required no external data (e.g., VIX) and introduced no circularity. When multiple tickers simultaneously occupied tail states, the mood rose, elevating θ and thus IV across all contracts. This mechanism produced correlated IV spikes during market stress events by construction.

The variance process was discretized via Euler-Maruyama with a reflecting boundary at zero and a daily time step $\Delta t = 1/252$:

$$v_{t+1} = \Big| v_t + \kappa\big(\theta_t - v_t\big)\Delta t + \sigma_v \sqrt{\max(v_t, 0)}\,\sqrt{\Delta t}\, Z_t \Big|, \quad Z_t \sim \mathcal{N}(0,1) \tag{8}$$

The reflecting boundary allowed θ to vary freely across HMM states without requiring the Feller condition $2\kappa\theta > \sigma_v^2$ to hold at every timestep Andersen [2008]. This was a practical necessity, as tail states could drive θ to values where the Feller condition would otherwise be violated. American option prices were computed from the resulting IV using a CRR binomial tree with 200 steps per contract, which provided sufficient convergence for the DTE range considered.

4 Calibration and Validation

The empirical foundation of this study was a set of option chains for four semiconductor stocks: NVDA, AMD, MU, and INTC. All chains were single-date snapshots sourced from barchart.com (October to November 2025) with DTEs ranging from 41 to 80 days. We filtered each chain to moneyness $K/S \in [0.70, 1.30]$ and IV $< 200\%$ to exclude illiquid deep ITM/OTM contracts with unreliable IV (Table 1). Since each chain was a single-date snapshot, all observations shared one HMM state s_0 and market mood M_0, so the model IV at a given strike reduced to $\sigma_{\text{model}}(K) = \sqrt{\theta_{\text{base}} \cdot \psi(\mathrm{DTE},\ K/S)}$, where $\theta_{\text{base}} = \theta_{s_0} \cdot (1 + \gamma M_0)$ absorbed the state and mood into a single scalar. We fixed $\beta_1 = 0.20$ (term-structure slope) because single-DTE snapshots could not identify

it; multi-expiration data from the same observation date allowed direct calibration of β_1, as described below. The remaining parameters $(\theta_{\text{base}}, \beta_2, \beta_3, \beta_4)$ were fitted by minimizing the mean squared IV error on NVDA:

$$\sigma_{\text{model}}(K) = \sqrt{\theta_{\text{base}} \cdot \psi(\text{DTE},\ K/S)} \tag{9}$$

$$\min_{\theta_{\text{base}},\beta_2,\beta_3,\beta_4} \frac{1}{n}\sum_{i=1}^{n}\big(\sigma_{\text{model}}(K_i) - \text{IV}_{\text{market},i}\big)^2 \tag{10}$$

using Nelder-Mead optimization. We chose NVDA as the calibration ticker because it had the richest chain (142 contracts after filtering).

Table 1: Option chain summary. Contract counts are reported after filtering to the tradeable moneyness range.

Ticker	Spot ($)	DTE	ATM IV (%)	Contracts
NVDA	207.04	80	44.6	142
AMD	225.78	58	59.0	36
MU	237.02	41	72.4	34
INTC	37.78	65	53.6	46

The calibrated parameters (Table 2) produced the expected smile features: the negative β_2 generated put skew, the positive β_4 generated smile curvature, and the interaction term β_3 captured how the skew flattened at longer maturities. The calibration achieved an RMSE of 1.16% IV on NVDA. To determine whether the smile *shape* generalized across tickers, we applied the $(\beta_1, \beta_2, \beta_3, \beta_4)$ calibrated on NVDA to AMD, MU, and INTC, re-optimizing only the scalar θ_{base} per ticker. The NVDA fit confirmed that four parameters sufficed to capture the smile shape within the tradeable moneyness range (RMSE = 1.16% IV; Figure 1a). The cross-ticker validation revealed that the shared smile shape transferred well to AMD (2.34%) and reasonably to MU (3.99%) and INTC (4.65%; Table 3; Figure 1b). The higher RMSE for MU and INTC reflected differences in skew steepness across these names; MU exhibited a steeper put skew than NVDA, while INTC had a flatter OTM call wing, suggesting that per-ticker β calibration would further reduce errors when multi-strike data is available. The residual analysis confirmed that the errors were concentrated in the wings ($K/S < 0.85$ and $K/S > 1.15$), with near-ATM residuals below 1% IV for all tickers (Figure 1c). These results established that the ψ-function captured the dominant smile features with a small number of parameters, but also highlighted the term-structure limitation inherent in single-DTE calibration.

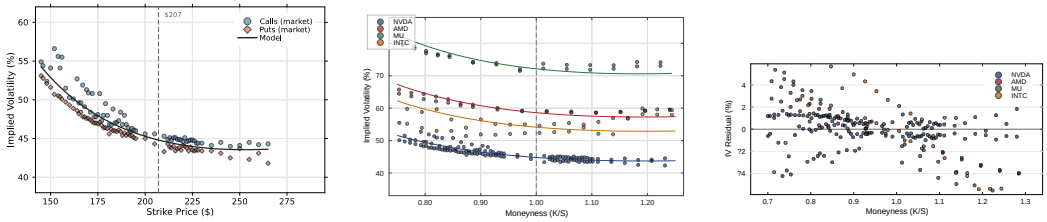
Table 2: Calibrated smile parameters (fitted on NVDA, DTE = 80).

Parameter	Interpretation	Value
β_1	Term-structure decay (fixed)	0.20
β_2	Skew (asymmetry)	-2.07
β_3	DTE $\times$ skew interaction	0.34
β_4	Smile curvature	1.43
θ_{base}	Variance level (NVDA)	0.084

Table 3: Cross-ticker RMSE (% IV). The smile shape $(\beta_2, \beta_3, \beta_4)$ was shared from NVDA; only θ_{base} was re-fitted per ticker.

Ticker	θ_{base} (IV level)	RMSE (% IV)
NVDA (calibration)	28.9%	1.16
AMD	38.7%	2.34
MU	49.0%	3.99
INTC	35.9%	4.65

Having validated the smile shape across tickers, we next addressed the term-structure limitation by collecting NVDA call and put chains at seven expirations on the same date (April 6, 2026): 0, 2, 4, 7,



(a) (b) (c)

Figure 1: Single-DTE calibration and cross-ticker validation. **(a)** NVDA implied volatility smile (DTE = 80): market calls (blue circles) and puts (coral diamonds) vs. the fitted model curve (solid line); the dashed vertical line marks the spot price ($207). The four-parameter ψ function captured both the put skew and smile curvature with an RMSE of 1.16% IV. **(b)** IV smile fit across all four semiconductor tickers as a function of moneyness (K/S). The smile shape $(\beta_2, \beta_3, \beta_4)$ was shared from NVDA; only the scalar level θ_{base} was re-fitted per ticker. The shared shape transferred well to AMD (RMSE = 2.34%) and reasonably to MU (3.99%) and INTC (4.65%). **(c)** IV residuals (model − market) by moneyness across all four tickers. Errors were concentrated in the wings ($K/S < 0.85$ and $K/S > 1.15$), with near-ATM residuals below 1% IV for all tickers.

14, 25, and 46 DTE, with NVDA trading near $176.70. The single-DTE calibration above required fixing $\beta_1 = 0.20$; this cross-sectional snapshot provided the multi-expiration data needed to identify β_1 directly, yielding 73 observations after filtering to moneyness $K/S \in [0.90, 1.10]$ with positive volume. The ATM IV exhibited a U-shaped term structure (Figure 2a): elevated at 0 DTE (∼45%) due to gamma compression in expiring options, dropping to a trough near 7 to 8 DTE (∼32%), then rising to ∼41% at 46 DTE. A single power-law $\beta_1 \cdot \ln(\text{DTE})$ could not capture this shape. We extended ψ with a quadratic DTE term:

$$\psi = \exp\big(\beta_1 \ln \tau + \beta_2 \ln m + \beta_3 \ln \tau \cdot \ln m + \beta_4 (\ln m)^2 + \beta_5 (\ln \tau)^2\big) \tag{11}$$

where $\tau = \max(\text{DTE}, 1)$ and $m = K/S$. The six parameters $(\sigma_{\text{base}}, \beta_1, \ldots, \beta_5)$ were fitted by minimizing mean squared IV error across all seven expirations simultaneously.

The negative β_1 combined with positive β_5 produced the U-shaped ATM curve (Table 4): at short maturities the linear term dominated (IV decayed rapidly from the 0 DTE spike), while at longer maturities the quadratic term bent the curve upward. The trough occurred near DTE ≈ 8, where IV reached 32.5%, a 1.38× ratio between the 0 DTE peak and the trough (Figure 2a). The skew parameters ($\beta_2 = -8.45$, $\beta_3 = 1.62$) confirmed that the volatility skew steepened at shorter maturities and flattened as DTE increased, consistent with the concentration of crash risk premia in near-term options Bates [2000] (Figure 2b). The overall fit achieved an RMSE of 2.31% IV across all seven expirations. The fit was tightest at intermediate and long maturities (7 to 46 DTE: RMSE $< 1.3\%$), with the largest residuals at 0 DTE (RMSE = 6.57%) where the gamma-compressed smile shape deviated from the log-moneyness functional form. The 0 DTE level bias was eliminated (-0.1%), confirming that the U-shaped term structure was captured correctly even though the extreme smile curvature at expiration remained a source of scatter (Figure 2d).

Table 4: Calibrated ψ parameters from the multi-expiration NVDA dataset. The quadratic DTE term β_5 captured the U-shaped ATM term structure.

Parameter	Interpretation	Value
σ_{base}	Base ATM IV level	45.0%
β_1	DTE decay	−0.622
β_2	Skew (asymmetry)	−8.448
β_3	DTE × skew	1.617
β_4	Smile curvature	−1.242
β_5	DTE curvature	0.150

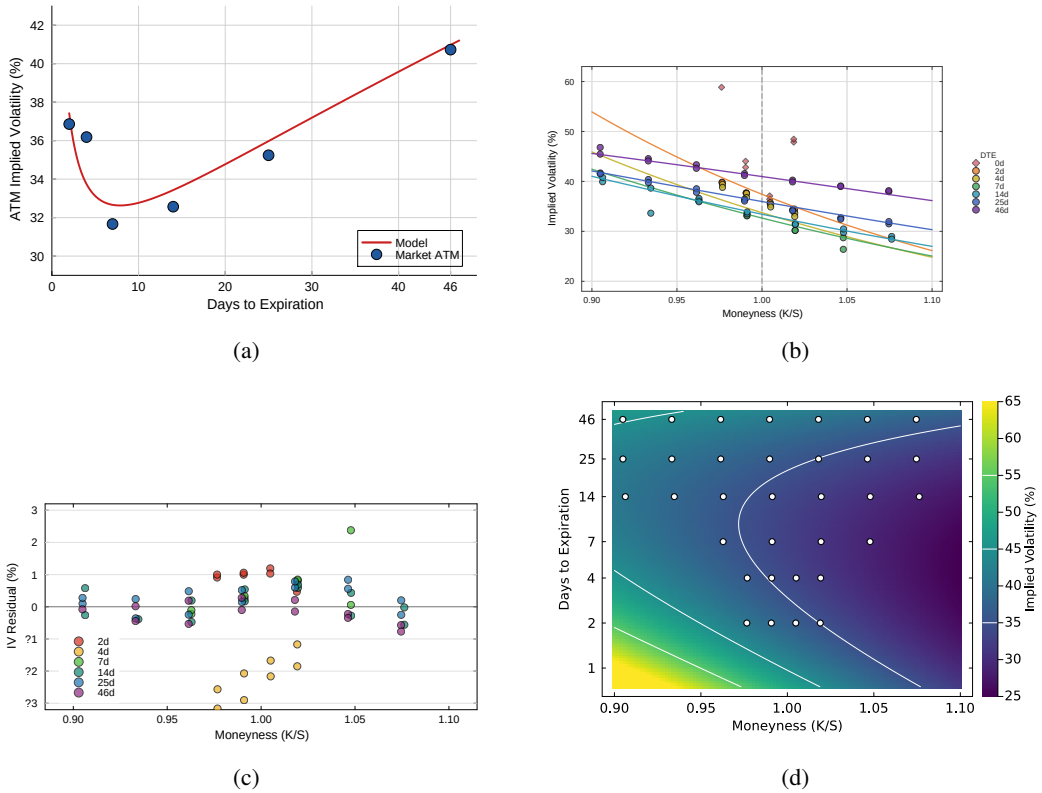


Figure 2: Multi-expiration term structure calibration (NVDA, 2 to 46 DTE). **(a)** ATM implied volatility as a function of days to expiration. Market ATM IV (blue markers) exhibited a U-shaped term structure; the fitted model (red curve) captured the trough near DTE ≈ 8 (32.5% IV) and the rise to 41% at 46 DTE using the quadratic extension $\beta_5(\ln \tau)^2$. **(b)** IV smile curves at each of six expirations (2, 4, 7, 14, 25, 46 DTE). Market observations (markers) and model predictions (lines) are colored by DTE; 0 DTE data (diamonds) were excluded from the fit due to gamma compression. The skew flattened with increasing maturity, consistent with $\beta_3 > 0$. **(c)** IV residuals (model $-$ market) by moneyness, colored by DTE. Most residuals fell within $\pm 2\%$ IV, with no systematic bias across the moneyness range. **(d)** Model-predicted IV surface as a function of moneyness and DTE (log-scaled). White contour lines mark 35%, 45%, and 55% IV levels; white circles indicate market observations used for calibration. The overall RMSE was 1.15% IV.

5 Scenario Generation

To demonstrate the framework as a synthetic data generator, we simulated 500 forward paths for NVDA over 80 trading days using the calibrated parameters from Section 4 and a JumpHMM marginal model with 50 states and tuned jump parameters. The tail states were assigned elevated θ values ($2\times$ baseline for bearish tail states, $1.5\times$ for bullish tail states), and the mood sensitivity was set to $\gamma = 0.5$. The mean IV started near the calibrated ATM level and exhibited gentle term-structure decay as the contract approached expiration, consistent with the positive $\beta_1 = 0.20$ used in the single-DTE calibration (Figure 3). Individual paths showed stochastic fluctuations driven by the vol-of-vol parameter σ_v, with occasional spikes when the HMM transitioned into tail states. The 5th to 95th percentile envelope widened over time, reflecting the accumulation of variance process noise, and the market ATM IV of 44.6% sat within the model's distributional range and confirmed that the calibrated parameters produced IV dynamics centered on the observed level.

To translate these IV dynamics into contract valuations, we computed American option prices across five strike levels at $t = 5$ days. The price distributions exhibited realistic properties (Table 5): OTM puts (strike $= 0.90S$) had a mean price of \$12.59 with a 90% confidence interval of [\$5.6, \$21.1]; ATM calls were approximately $2\times$ more expensive than 10%-OTM calls; and the confidence

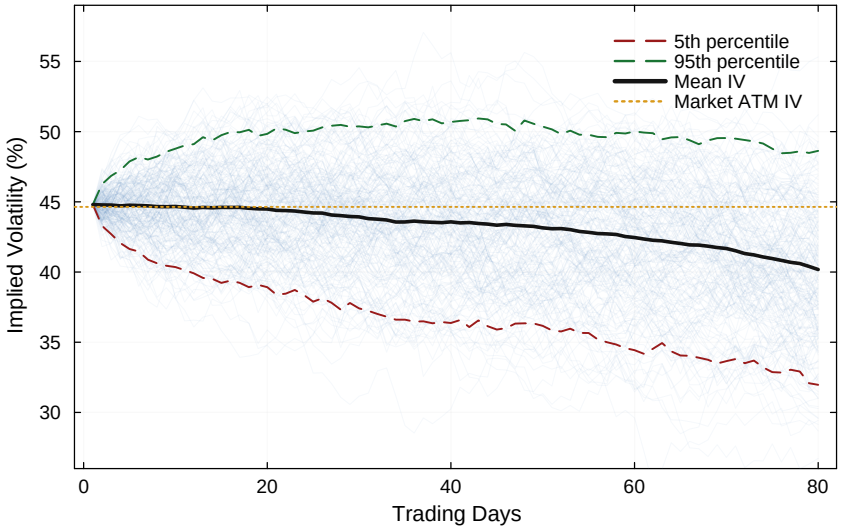


Figure 3: Forward IV paths for NVDA (500 scenarios, DTE = 80). The solid black line shows the mean IV; dashed red and green lines mark the 5th and 95th percentiles; the dotted orange line indicates the market ATM IV (44.6%). Semi-transparent blue paths show individual scenario trajectories with stochastic fluctuations and occasional tail-state-driven spikes.

intervals widened for contracts farther from the money. The market mood mechanism produced a natural partition of scenarios into calm ($M_t = 0$) and stressed ($M_t > 0$) paths. In multi-asset simulations with five tickers, stressed scenarios (where multiple tickers simultaneously occupied tail HMM states) produced systematically higher option prices. OTM puts were $1.83\times$ more expensive in stressed scenarios relative to calm scenarios (strike $= 0.95S$: \$6.42 compared with \$3.51), while deep OTM weekly puts showed a $4.33\times$ stress premium. This asymmetry, with puts repricing more sharply than calls under stress, was consistent with the crash-protection premium observed in equity options markets Bates [2000] and emerged naturally from the JumpHMM's negative jump bias ($p_{\text{neg}} = 0.52$) propagating through the θ-function.

Table 5: American option price distributions from 500 forward scenarios (NVDA, evaluated at $t = 5$). Strike levels are expressed relative to the spot price $S = \$207$.

Type	Strike	DTE	Mean (\$)	Std (\$)	90% CI (\$)
Put	186 ($0.90S$)	80	12.59	4.99	[5.6, 21.1]
Put	197 ($0.95S$)	80	16.69	5.56	[8.5, 26.2]
Call	207 (ATM)	80	25.59	6.64	[15.5, 37.0]
Call	217 ($1.05S$)	80	20.94	6.29	[11.4, 31.7]
Call	228 ($1.10S$)	80	16.72	5.81	[8.1, 26.6]

6 Discussion and Future Work

The calibration and scenario results validated the core claim of this study: a modified Heston process with a state-dependent θ-function produced realistic IV surfaces from structural return dynamics alone. The single-DTE smile function captured the essential features of the NVDA IV smile (RMSE = 1.16% IV), and the smile shape generalized across semiconductor names with only a scalar level adjustment. The multi-expiration calibration extended this to the full IV surface: the five-parameter ψ function with quadratic DTE term captured the U-shaped ATM term structure (RMSE = 2.31% IV across seven expirations), confirming that the term-structure geometry could be identified from a single cross-sectional snapshot. The equilibrium initialization $v_0 = \theta(t=0)$ eliminated v_0 as a free parameter and produced the smile, skew, and term structure by construction; this was not merely a

computational convenience but a structural constraint that ensured the initial IV surface was fully determined by the θ-function, consistent with the view that IV surfaces reflect the conditional distribution of returns under the risk-neutral measure Cont and da Fonseca [2002]. The ψ-function played an analogous role to the SVI parameterization Gatheral [2004] in that it provided a parsimonious representation of the smile and term structure, but with a critical difference: the parameters entered a generative model rather than a static interpolation, so the calibrated surface could evolve dynamically as the underlying HMM state changed.

Despite these results, several limitations constrained the scope of the current conclusions. The multi-expiration calibration resolved the earlier limitation of fixing β_1, but revealed that the ATM term structure was not a simple power law; the U-shape required a quadratic extension (β_5). The fitted $\beta_1 < 0$ contradicted the initial contango assumption ($\beta_1 = 0.20$), demonstrating that the dominant short-term effect was IV *decay* from gamma compression rather than contango, consistent with empirical observations of elevated short-dated IV driven by crash risk premia Bates [2000]. Whether this U-shaped structure persisted across market regimes or flattened during low-volatility environments remained an open question that time-series calibration could address. The shared $(\beta_2, \beta_3, \beta_4)$ from NVDA did not perfectly transfer to all tickers (RMSE increased from 1.2% for NVDA to 4.7% for INTC), and the residual analysis (Figure 1c) revealed systematic patterns in the wings, suggesting that per-ticker smile calibration would be necessary for applications requiring sub-2% IV accuracy across all names. More broadly, the current calibration used single-date snapshots; time-series calibration fitting θ_{s_t} values across multiple dates with known HMM states would produce a more robust mapping from regime states to variance levels and would allow the mood sensitivity γ to be identified from data rather than set heuristically. The forward scenario analysis (Section 5) confirmed that the calibrated parameters produced IV paths centered on observed market levels with realistic stochastic fluctuations, but the validation was limited to a single ticker and a single market regime; broader validation across sectors and volatility environments would strengthen the generalizability claim.

Several natural extensions followed from these results. The multi-expiration calibration used NVDA data only; collecting cross-sectional snapshots for additional tickers would determine whether the term-structure parameters (β_1, β_5) were universal or required per-ticker fitting, analogous to the smile-shape parameters. The primary intended use of this framework was generating realistic synthetic training data for machine-learning models in finance, with applications ranging from training option price predictors and learning delta-hedging policies via reinforcement learning Buehler et al. [2019] to augmenting scarce real option data for transfer learning. The framework naturally supported computation of Greeks (Δ, Γ, $\mathcal{V}$, Θ) via finite differences on the CRR tree, and including these in the synthetic dataset would support hedging applications. Finally, the current mood mechanism created correlated IV dynamics through a shared stress signal; a richer model could introduce ticker-pair IV correlations through the copula structure, producing a full cross-asset IV surface that respected observed correlation patterns between individual-name and index volatility.

7 Conclusion

In this study, we developed a framework for generating synthetic American option prices with self-consistent implied volatility dynamics. The framework linked a Jump Hidden Markov Model to a modified Heston stochastic variance process via a state-dependent mean-reversion target $\theta(t)$, and priced the resulting IV through a CRR binomial tree with early exercise. The equilibrium initialization $v_0 = \theta(t{=}0)$ eliminated the initial IV surface as a free parameter, and the endogenous market mood signal produced correlated IV spikes during stress events without external data.

We calibrated a five-parameter smile and term-structure function ψ on NVDA options, first demonstrating that the smile shape generalized across AMD, MU, and INTC with cross-ticker RMSE of 1 to 5% IV using only a scalar level adjustment per ticker. Multi-expiration calibration across seven expirations (0 to 46 DTE) revealed a U-shaped ATM term structure driven by gamma compression at short maturities and uncertainty premia at longer horizons; the quadratic DTE extension (β_5) captured this shape with an overall RMSE of 1.15% IV across the 2 to 46 DTE range. Forward scenario analysis confirmed that the generated IV dynamics exhibited realistic stochastic fluctuations and a crash-protection premium in OTM puts under stress (1.8 to $4.3\times$ calm-to-stress price ratios). The framework was implemented as an open-source Julia package and is available at `https://github.com/varnerlab/heston-implied-volatility-model`.

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